全民英檢中級 GEPT Intermediate | 第 22 單元 | 科學與度量 · Science & Measurement

Students Catch a Red Moon from the School Roof

學生在學校屋頂捕捉紅月亮

TAIPEI - More than eighty students stayed at school until midnight last Saturday, but not to study for exams. The astronomy club of Linkou Senior High School had set up three telescopes on the roof so that members could watch a total lunar eclipse, a phenomenon that will not be visible again from Taiwan for another two years.

According to club adviser Mr. Chen, a former physicist who now teaches earth science, the eclipse happens when the Earth moves directly between the Sun and the Moon. 'The Moon does not produce its own light; it only reflects solar rays,' he explained to the crowd. 'When our planet blocks that light, the Moon slowly turns a deep red.' Students recorded the color every ten minutes, and their observation sheets will be sent to a research team at National Central University.

For many members, the night was their first chance to use scientific equipment outside a classroom. Second-year student Amy Wu said she had always believed the theory that the Moon's orbit was a perfect circle. 'Mr. Chen showed us that it is actually slightly oval, and that is why the Moon sometimes looks bigger,' she said. Another student attached a smartphone to the telescope to photograph the surface structure of the Moon, including craters several hundred meters deep.

The club plans to hold a similar event in December, when a spacecraft launched by Japan is expected to pass close to the Moon. Mr. Chen hopes the activities will stimulate more interest in science. 'A cold roof at midnight may not sound exciting,' he said, 'but once you have seen the sky through a telescope, you never look at it the same way again.'

重點單字 Key Vocabulary

單字	詞性	中文	例句
lunar	*adj.*	月亮的	The Lunar New Year is the most important holiday in Taiwan. 農曆新年是臺灣最重要的節日。

telescope	*n.*	望遠鏡	With a telescope, you can see the rings of Saturn. 用望遠鏡可以看到土星的環。
phenomenon	*n.*	現象	A rainbow is a natural phenomenon caused by sunlight and rain. 彩虹是由陽光和雨水造成的自然現象。
physicist	*n.*	物理學家	The physicist explained how a roller coaster stays on its track. 那位物理學家解釋雲霄飛車如何停留在軌道上。
reflect	*v.*	反射	The calm lake reflected the mountains perfectly. 平靜的湖面完美地倒映出群山。
solar	*adj.*	太陽的	Many houses in Tainan now use solar panels on their roofs. 臺南現在有很多房子在屋頂裝太陽能板。
observation	*n.*	觀察；觀測	Careful observation helped the nurse notice the problem early. 仔細的觀察幫助護理師及早發現問題。
orbit	*n.*	軌道	The satellite will stay in orbit for about ten years. 這顆衛星將在軌道上停留約十年。
structure	*n.*	結構；構造	The structure of the bridge was checked after the earthquake. 地震後，橋梁的結構接受了檢查。
stimulate	*v.*	刺激；激發	Reading stories can stimulate a child's imagination. 閱讀故事可以激發孩子的想像力。

片語 Phrases

set up	架設；設立	The volunteers set up tents before the typhoon arrived. 志工們在颱風來臨前架好了帳篷。
according to	根據	According to the news, the MRT will close early tonight. 根據新聞報導，捷運今晚會提早收班。
the same way	同樣的方式	After the trip, I never looked at the ocean the same way. 那次旅行之後，我再也不用同樣的眼光看海洋了。

文法與句型 Grammar

S + V ... so that + S + can / could + V

so that 引導目的副詞子句，表示「以便、為了讓…」，子句中常搭配 can、could、will、would 等助動詞。文中 had set up three telescopes so that members could watch 說明架設望遠鏡的目的。

She saved money so that she could buy a new bike.

她存錢是為了能買一輛新腳踏車。

once + S + V, S + V

once 當連接詞時意思是「一旦…就…」，引導時間或條件副詞子句，後面的主要子句說明結果。文中 once you have seen the sky through a telescope, you never look at it the same way 即為此用法。

Once you learn to ride a bike, you never forget how.

一旦你學會騎腳踏車，就永遠不會忘記。

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